



2013

YEARLY EXAMINATION

Preliminary Mathematics

General Instructions

Reading Time – 5 minutes

Working Time – 2 hours

Write using black or blue pen

Board-approved calculators may be used

All necessary working should be shown in every question.

Total Marks – 84

Attempt Questions 1 – 7

All questions are of equal value.

At the end of the examination, place your solution booklets in order and place them inside your question paper.

Submit ONE bundle.

Student Name: _____ **Student Number:** _____

Teacher: _____

QUESTION	MARK
1	/12
2	/12
3	/12
4	/12
5	/12
6	/12
7	/12
TOTAL	/84

Total Marks – 84

Attempt Questions 1 – 7

Questions are of equal value.

Begin each question in a SEPARATE writing booklet. Extra writing booklets are available.

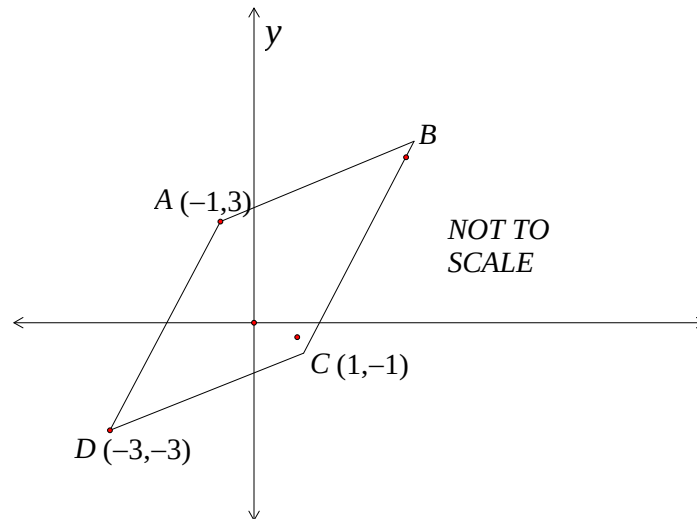
Question 1 (12 Marks) Use a SEPARATE writing booklet.

- (a) Evaluate $\sqrt[3]{\frac{1.2 \times 10^4 - 3.3 \times 10^2}{2.48}}$ correct to three significant figures. 2
- (b) Find a and b such that $(2 - 5\sqrt{3})^2 = a + b\sqrt{3}$. 2
- (c) Write $\frac{3-q}{3} - \frac{2+q}{4}$ as a single fraction in simplest form. 2
- (d) Write $\frac{8^x - 18^x}{12^x - 27^x}$ in simplest form. 2
- (e) Solve $3x^2 - 7x = 11$. Leave your answer in exact form. 2
- (f) Solve $|x + 2| \geq 5$. 2

Question 2 (12 Marks) Use a SEPARATE writing booklet.

(a) Fully factorise $8x^3 - 27$. 2

(b) In the diagram below, $ABCD$ is a parallelogram and the points A , C and D , are $(-1, 3)$, $(1, -1)$ and $(-3, -3)$ respectively.



(i) Find the length of the interval DC . 1

(ii) Show that the equation of the line passing through D and C is $x - 2y - 3 = 0$. 2

(iii) Determine the angle of inclination that the line through DC makes with x -axis to the nearest minute.. 1

(iv) Find the midpoint of AC . 1

(v) Hence or otherwise, find the co-ordinates of B . 2

(vi) Find the perpendicular distance from A to DC . 2

(vii) Find the area of the parallelogram $DABC$. 1

Question 3 (12 Marks) Use a SEPARATE writing booklet.

(a) Differentiate with respect to x :

(i) $2x^3 - 7x + 5$. 1

(ii) $\frac{5}{x^3} - 2\sqrt{x}$. 2

(iii) $(1 - 8x^2)^5$. 2

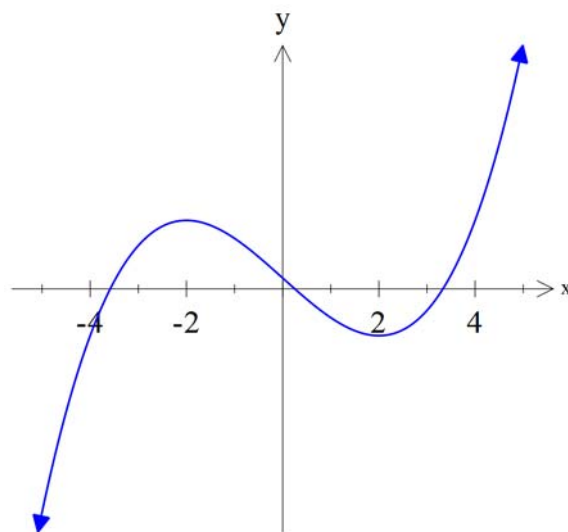
(b) (i) Given $f(x) = (x+1)(2+3x)^6$ find $f'(x)$. 2

(ii) Find the values for x for which the tangent is horizontal. 2

(c) Find the equation of the normal to the curve $y = 2x^3 + 4x - 5$ at the point where $x = 1$. 3

Question 4 (12 Marks) Use a SEPARATE writing booklet.

- (a) Below is the graph of $y = f(x)$.



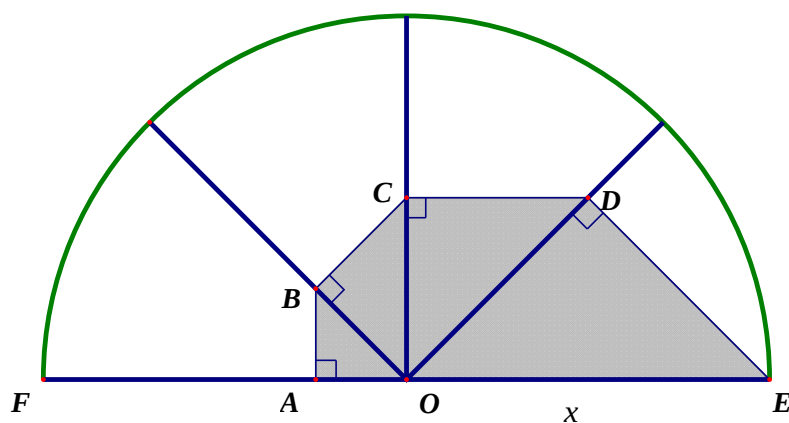
Sketch the graph of $y = f'(x)$.

2

- (b) Solve $2 \cos^2 \theta - 1 = 0$ for $0^\circ \leq \theta \leq 360^\circ$.

3

- (c) The diagram below shows a semicircle with radius x . The semicircle is divided into four equal sectors. $ED \perp OD$, $DC \perp OC$, $CB \perp OB$ and $BA \perp OA$.



- (i) Explain briefly why $\triangle ODE$ is an isosceles triangle. 1
- (ii) Calculate the exact length of OD in terms of x . 1
- (iii) Prove that $\triangle ODE \parallel \triangle OCD$. 2
- (iv) Calculate the ratio of the area of $\triangle OAB$ to the area of the shaded region $ABCDE$. 3

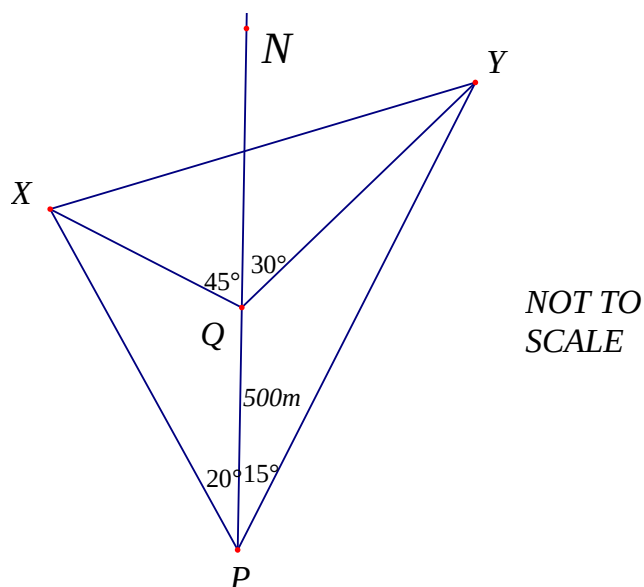
Question 5 (12 Marks) Use a SEPARATE writing booklet.

- (a) Sketch the graph of $y = f(x)$, where $f(x)$ is defined below.

4

$$f(x) = \begin{cases} \frac{1}{x}, & \text{for } x < -2 \\ x+1, & \text{for } -2 \leq x < 0 \\ 4-x^2, & \text{for } x \geq 0 \end{cases}$$

- (b) A sailor at point P observes two buoys at X and Y on a bearing of 340° and 015° respectively.
He sails due North for 500 m to a point Q and then observes X and Y on a bearing of 315° and 030° .



- (i) Show that $XP = \frac{250\sqrt{2}}{\sin 25^\circ}$. 2
- (ii) Show that $YP = \frac{250}{\sin 15^\circ}$. 2
- (iii) Find the distance XY to the nearest metre. 2
- (iv) Find the bearing of X from Y . 2

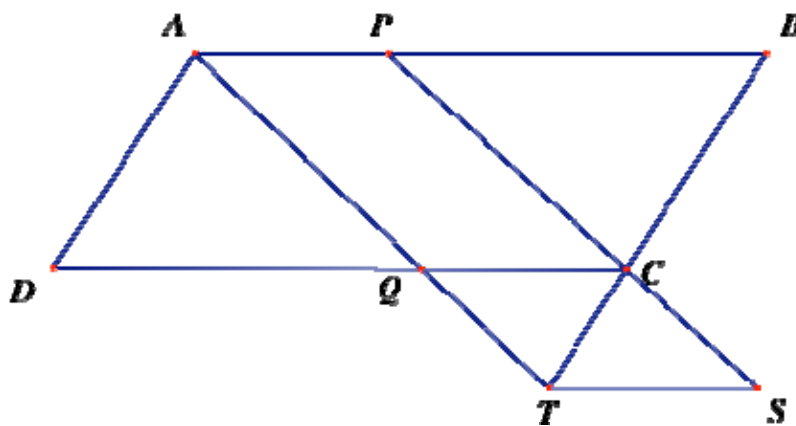
Question 6 (12 Marks) Use a SEPARATE writing booklet.

- (a) Find the points of intersection of $2x - y - 5 = 0$ and $y = \frac{9}{x-3} - 6$ 3

- (b) Sketch the region of the number plane where the following hold simultaneously: 3

$$(x-5)^2 + y^2 \leq 25, \quad y < x, \quad \text{and} \quad y \geq 0$$

- (c) In the following diagram $ABCD$ is a parallelogram.
 $AD = AQ = PC$. AQ and BC are produced to intersect at T .
 PC is produced to S such that $\angle PST = \angle SPB$.



- (i) Prove that $\triangle DAQ \equiv \triangle BCP$. 3
- (ii) Show that $\angle AQD = \angle PCQ$. 1
- (iii) Hence, prove that $APST$ is a parallelogram. 2

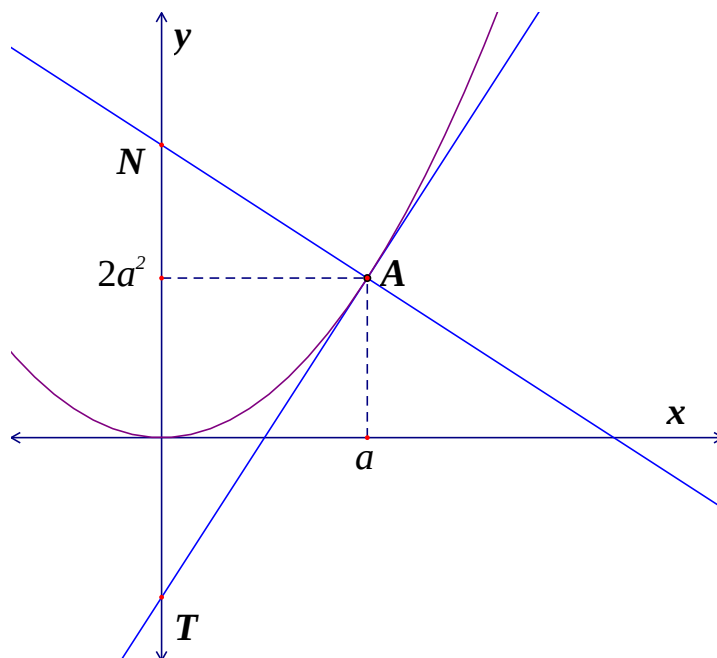
Question 7 is on the next page

Question 7 (12 Marks) Use a SEPARATE writing booklet.

(a) (i) Prove that $\frac{\tan x}{\sec x + 1} + \frac{\tan x}{\sec x - 1} = 2 \operatorname{cosec} x$. 2

(ii) Hence solve $\frac{\tan x}{\sec x + 1} + \frac{\tan x}{\sec x - 1} = 4$ for $0 \leq x \leq 360^\circ$. 2

- (b) In the diagram below, the function $y = 2x^2$ has been drawn with a point $A(a, 2a^2)$ such that $a > 0$.
The tangent at the point $A(a, 2a^2)$ intersects the y -axis at T and the normal at A intersects the y -axis at N .



(i) Show that the point T has coordinates $(0, -2a^2)$. 2

(ii) Show that the point N has coordinates $\left(0, \frac{1+8a^2}{4}\right)$. 2

(iii) Show that the area of $\triangle TAN$ is equal to $\frac{a}{8} + 2a^3$. 2

(iv) Find the area of $\triangle TAN$ when it is an isosceles triangle. 2

End of paper.

QUESTION 1

a)

$$\sqrt[3]{\frac{1.2 \times 10^4 - 3.3 \times 10^2}{2.48}}$$

$$= 16.75739056$$

$$= 16.8$$

b)

$$(2 - 5\sqrt{3})^2$$

$$= 2^2 - 2 \times 2 \times 5\sqrt{3} + (5\sqrt{3})^2$$

$$= 4 - 20\sqrt{3} + 75$$

$$= 79 - 20\sqrt{3}$$

$$\therefore a = 79 \quad b = -20$$

c)

$$\frac{3-q}{3} - \frac{2+q}{4}$$

$$\frac{4(3-q) - 3(2+q)}{12}$$

$$= \frac{12 - 4q - 6 - 3q}{12}$$

$$= \frac{6 - 7q}{12}$$

d)

$$\frac{8^x - 18^x}{12^x - 27^x}$$

$$\frac{2^{3x} - (3^{2x})(2^x)}{(2^{2x})(3^x) - (3^{3x})}$$

$$\frac{(2^x)(2^{2x} - 3^{2x})}{(3^x)(2^{2x} - 3^{2x})} = \frac{2^x}{3^x}$$

e)

$$3x^2 - 7x = 11$$

$$3x^2 - 7x - 11 = 0$$

$$x = \frac{7 \pm \sqrt{(-7)^2 - 4(3)(-11)}}{2(3)}$$

$$x = \frac{7 \pm \sqrt{49 + 132}}{6}$$

$$x = \frac{7 \pm \sqrt{181}}{6}$$

f)

$$|x + 2| \geq 5$$

$$x + 2 \geq 5 \quad \text{OR} \quad -(x + 2) \geq 5$$

$$x \geq 3 \quad \quad \quad -(x + 2) \geq 5$$

$$-x - 2 \geq 5$$

$$-x \geq 7$$

$$x \leq -7$$

QUESTION 2

a)

$$8x^3 - 27$$

$$(2x - 3)(4x^2 + 6x + 9)$$

b)i)

$$d_{DC} = \sqrt{(-3 - -1)^2 + (-3 - 1)^2}$$

$$d_{DC} = \sqrt{4 + 16} = \sqrt{20} = 2\sqrt{5}$$

b)ii)

$$m_{DC} = \frac{(-3 - -1)}{(-3 - 1)} = \frac{-2}{-4} = \frac{1}{2}$$

Equation of DC

$$y - (-1) = \frac{1}{2}(x - 1)$$

$$2y + 2 = x - 1$$

$$0 = x - 2y - 3$$

b)iii)

Let θ be angle of inclination.

$$\tan \theta = m_{DC} = \frac{1}{2}$$

$$\tan^{-1} \frac{1}{2} = \theta$$

$$\theta = 26.56505188$$

$$\theta = 26^\circ 34'$$

b)iv)

Midpoint_{AC}

$$\left(\frac{-1+1}{2}, \frac{3+ -1}{2} \right)$$

Midpoint is (0,1)

b)v)

Midpoint_{AC} = Midpoint_{DB}

$$0 = \frac{-3 + x_B}{2}$$

$$x_B = 3$$

$$1 = \frac{-3 + y_B}{2}$$

$$y_B = 5$$

Co-ordinates of B are (3,5)

b)vi)

Equation of DC

$$0 = x - 2y - 3$$

Point A(-1,3)

$$d = \frac{|(1)(-1) + (-2)(3) + (-3)|}{\sqrt{(1)^2 + (-2)^2}}$$

$$d = \frac{|-1 - 6 - 3|}{\sqrt{5}} = \frac{10}{\sqrt{5}}$$

b)vii)

Area_{PARALLELOGRAM} = Perp Height $\times$ Base

$$Area_{ABCD} = DC \times \frac{10}{\sqrt{5}}$$

$$= 2\sqrt{5} \times \frac{10}{\sqrt{5}} = 20 \text{ units}^2$$

QUESTION 3

a)

$$i) \frac{d}{dx} 2x^3 - 7x + 5 = 6x^2 - 7$$

$$ii) \frac{d}{dx} \left(\frac{5}{x^3} - 2\sqrt{x} \right) = \frac{d}{dx} \left(5x^{-3} - 2x^{\frac{1}{2}} \right)$$

$$= -15x^{-4} - x^{-\frac{1}{2}}$$

$$= -\frac{15}{x^4} - \frac{1}{\sqrt{x}}$$

$$iii) \frac{d}{dx} (1 - 8x^2)^5 = 5(1 - 8x^2)^4 (-16x)$$

$$= -80x(1 - 8x^2)^4$$

b)i)

$$f(x) = (x+1)(2+3x)^6$$

$$f'(x) = u'v + v'u$$

$$f'(x) = 1 \times (2+3x)^6 + 6(2+3x)^5 \times 3(x+1)$$

$$f'(x) = (2+3x)^6 + 18(2+3x)^5(x+1)$$

ii)

Horizontal tangent when $f'(x) = 0$

$$0 = (2+3x)^5 [(2+3x) + 18(x+1)]$$

$$0 = (2+3x)^5 (20+21x)$$

$$x = -\frac{2}{3} \quad x = -\frac{20}{21}$$

c)

$$y = 2x^3 + 4x - 5$$

$$\frac{dy}{dx} = 6x^2 + 4$$

$$At x = 1$$

$$\frac{dy}{dx} = 6(1)^2 + 4$$

$$= 10$$

$$\therefore m_{Tangent} = 10 \text{ when } x = 1$$

$$\therefore m_{Normal} = \frac{-1}{10} \text{ when } x = 1$$

$$\text{and } y = 2(1)^3 + 4(1) - 5 = 1$$

Equation of Normal

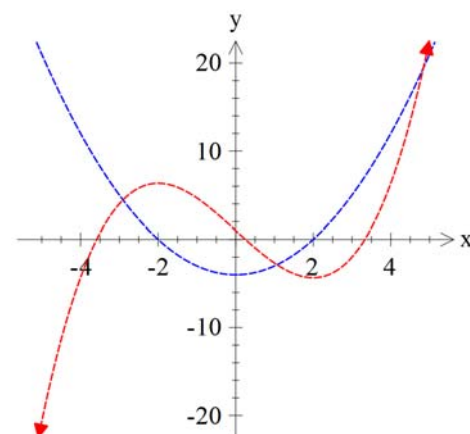
$$y - 1 = -\frac{1}{10}(x - 1)$$

$$10y - 10 = -x + 1$$

$$x + 10y - 11 = 0$$

QUESTION 4

a)



b)

$$2\cos^2 \theta - 1 = 0 \text{ for } 0^\circ \leq \theta \leq 360^\circ.$$

$$\cos^2 \theta = \frac{1}{2}$$

$$\cos \theta = \pm \frac{1}{\sqrt{2}}$$

Related angle

$$\cos^{-1} \frac{1}{\sqrt{2}} = 45^\circ$$

Solutions are in all four quadrants

$$x = 45^\circ$$

$$x = 180^\circ - 45^\circ = 135^\circ$$

$$x = 180^\circ + 45^\circ = 225^\circ$$

$$x = 360^\circ - 45^\circ = 315^\circ$$

c)i)

The semi circle is divided into four equal sectors so

$$\angle EOD = 180 \div 4 = 45^\circ$$

$$\angle EOD + \angle EDO + \angle DEO = 180^\circ$$

(Angle sum of $\triangle ODE$)

$$\therefore \angle DEO = 45^\circ$$

$\therefore \triangle ODE$ is an isosceles triangle

(Two equal angles)

c)ii)

$$x^2 = (OD)^2 + (ED)^2$$

$$x^2 = (OD)^2 + (OD)^2$$

$$x^2 = 2(OD)^2$$

$$\frac{x^2}{2} = (OD)^2$$

$$\frac{x}{\sqrt{2}} = OD$$

c)iii)

$$\angle EOD = \angle DOE = 180 \div 4 = 45^\circ$$

$\left(\begin{array}{l} \text{The semi circle is divided} \\ \text{into four equal sectors} \end{array} \right)$

$$\angle EDO = \angle DCO = 90^\circ$$

($DC \perp CO$ and $ED \perp DO$)

$\therefore \angle EDO \parallel \angle DCO$ (equiangular)

c)iv)

$$\frac{x^2}{2} = 2(OC)^2$$

$$\frac{x^2}{4} = (OC)^2$$

$$\frac{x}{2} = OC$$

$$\text{Similarly } \frac{x}{2\sqrt{2}} = OB \text{ and } \frac{x}{4} = OA$$

Area of $\triangle ABO$

$$\frac{1}{2} \left(\frac{x}{4} \right)^2 = \frac{x^2}{32}$$

Area of $ABCDE$

$$= \frac{1}{2} \left(\left(\frac{x}{4} \right)^2 + \left(\frac{x}{2\sqrt{2}} \right)^2 + \left(\frac{x}{2} \right)^2 + \left(\frac{x}{\sqrt{2}} \right)^2 \right)$$

$$= \frac{1}{2} \left(\left(\frac{x^2}{16} \right) + \left(\frac{x^2}{8} \right) + \left(\frac{x^2}{4} \right) + \left(\frac{x^2}{2} \right) \right)$$

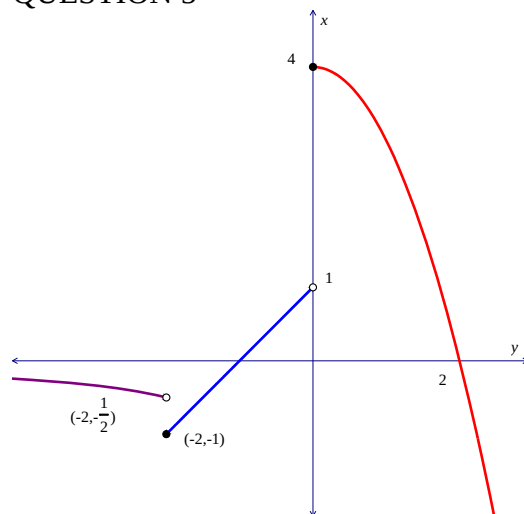
$$= \frac{x^2 + 2x^2 + 4x^2 + 8x^2}{32} = \frac{15x^2}{32}$$

Area of ABO : Area of $ABCDE$

$$\frac{x^2}{32} : \frac{15x^2}{32}$$

$$1:15$$

QUESTION 5



b)i)

$$\angle XQP = 180^\circ - 45^\circ = 135^\circ$$

$$\angle QXP = 180^\circ - 135^\circ - 20^\circ = 25^\circ$$

Using the sine rule in $\triangle QPY$

$$\frac{XP}{\sin 135^\circ} = \frac{500}{\sin 25^\circ}$$

$$XP = \frac{500}{\sin 25^\circ} \times \frac{1}{\sqrt{2}}$$

$$XP = \frac{250\sqrt{2}}{\sin 25^\circ}$$

bii)

$$\angle YQP = 180^\circ - 30^\circ = 150^\circ$$

$$\angle QYP = 180^\circ - 150^\circ - 15^\circ = 15^\circ$$

Using the sine rule in $\triangle QPY$

$$\frac{YP}{\sin 150^\circ} = \frac{500}{\sin 25^\circ}$$

$$YP = \frac{500}{\sin 25^\circ} \times \frac{1}{2}$$

$$YP = \frac{250}{\sin 25^\circ}$$

b)iii)

$$\angle YPX = 15^\circ + 20^\circ = 35^\circ$$

Using the cosine rule in $\triangle XPY$

$$(YX)^2 = (YP)^2 + (XP)^2 - 2(XP)(YP)\cos(\angle XPY)$$

$$(YX)^2 = \left(\frac{250}{\sin 15^\circ}\right)^2 + \left(\frac{250\sqrt{2}}{\sin 25^\circ}\right)^2 - 2\left(\frac{250}{\sin 15^\circ}\right)\left(\frac{250\sqrt{2}}{\sin 25^\circ}\right)\cos(35^\circ)$$

$$(YX)^2 = \left(\frac{250}{\sin 15^\circ}\right)^2 + \left(\frac{250\sqrt{2}}{\sin 25^\circ}\right)^2 - 2\left(\frac{250}{\sin 15^\circ}\right)\left(\frac{250\sqrt{2}}{\sin 25^\circ}\right)\cos(35^\circ)$$

$$(YX)^2 = 309007.3595$$

$$YX = 555.884304m$$

$$YX \approx 556m$$

b)iv)

Construct $N_Y S_Y$ through Y such that

$$N_Y S_Y \parallel PN$$

$$\angle NPY = \angle PYS_Y = 15^\circ (\text{alternate angles})$$

$$\text{Bearing}_{\text{of } X \text{ from } Y} = 180^\circ + 15^\circ + \angle PYX$$

Using the cosine rule in $\triangle XPY$

$$\frac{(PX)^2 - (XY)^2 - (PY)^2}{2(XY)(PY)} = \cos(\angle PYX)$$

$$\frac{\left(\frac{250\sqrt{2}}{\sin 25^\circ}\right)^2 - (556)^2 - \left(\frac{250}{\sin 15^\circ}\right)^2}{2(556)\left(\frac{250}{\sin 15^\circ}\right)} = \cos(\angle PYX)$$

$$\angle PYX = 59^\circ 40'$$

$$\begin{aligned} \text{Bearing}_{\text{of } X \text{ from } Y} &= 180^\circ + 15^\circ + 59^\circ 40' \\ &= 254^\circ 40' \end{aligned}$$

a)

$$y = \frac{9}{x-3} - 6 \text{ and } 2x - y - 5 = 0$$

$$\frac{9}{x-3} - 6 = 2x - 5$$

$$\frac{9}{x-3} = 2x + 1$$

$$9 = (2x + 1)(x - 3)$$

$$9 = 2x^2 - 5x - 3$$

$$0 = 2x^2 - 5x - 12$$

$$0 = (2x + 3)(x - 4)$$

$$x = 4 \quad x = -\frac{3}{2}$$

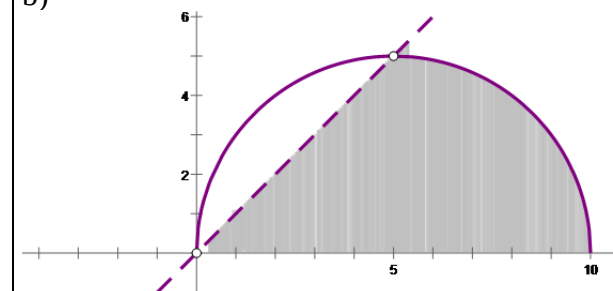
$$y = \frac{9}{4-3} - 6 \quad y = \frac{9}{-\frac{3}{2}-3} - 6$$

$$y = 3 \quad y = -8$$

Points of intersection

$$(4, 3) \text{ and } \left(-\frac{3}{2}, 6\right)$$

b)



i

In $\triangle ADQ$ and $\triangle BCP$

$AD = BC$ (Opposite sides in quadrilateral)

$\therefore AD = AQ = PC = BC$

$\angle ADQ = \angle CBP$ (Opposite angles in quadrilateral)

$\therefore \angle ADQ = \angle AQD$ (Equal angles opposite equal sides)

$\angle PBC = \angle BPC$ (Equal angles opposite equal sides)

$\therefore \angle BPC = \angle AQD$

$\therefore \triangle ADQ \equiv \triangle BCP$ (AAS)

ii

$AD = BC$ (Opposite sides in quadrilateral)

$\angle BPC = \angle AQD$ (From part i)

$AB \parallel DC$ (opposite side of quadrilateral)

$\angle BPC = \angle PCQ$ (Alternate angles in parallel lines.)

$\therefore \angle AQD = \angle PCQ$

iii)

$\angle AQD = \angle PCQ$ (Part ii)

$\therefore AT \parallel PS$ (Corresponding angles are equal)

$\angle TSP = \angle BPC$ (Given)

$\therefore AB \parallel TS$ (Alternate angles are equal)

$\therefore APST$ is a parallelogram (Opposite sides are parallel)

QUESTION 7

a)i

$$\frac{\tan x}{\sec x + 1} + \frac{\tan x}{\sec x - 1} = 2 \operatorname{cosec} x$$

$$\begin{aligned} LHS &= \frac{\tan x}{\sec x + 1} + \frac{\tan x}{\sec x - 1} \\ &= \frac{\tan x (\sec x + 1) + \tan x (\sec x - 1)}{\sec^2 x - 1} \end{aligned}$$

$$= \frac{2 \sec x \tan x}{\tan^2 x}$$

$$= \frac{2 \sec x}{\tan x} = 2 \times \frac{1}{\cos x} \times \frac{\cos x}{\sin x}$$

$$= \frac{2}{\sin x} = 2 \operatorname{cosec} x = RHS$$

a)ii)

$$\frac{\tan x}{\sec x + 1} + \frac{\tan x}{\sec x - 1} = 4$$

$$2 \operatorname{cosec} x = 4$$

$$\operatorname{cosec} x = 2$$

$$\sin x = \frac{1}{2}$$

x lies in the first and
second quadrant

$$x = 30^\circ$$

$$x = 150^\circ$$

b)i)

$$y = 2x^2$$

$$y' = 4x$$

When $x = a$ $y' = 4a$

Equation of tangent:

$$y - 2a^2 = 4a(x - a)$$

When $x = 0$

$$y - 2a^2 = 4a(0 - a)$$

$$y = 2a^2 - 4a^2 = -2a^2$$

Co-ordinates of T are $(0, -2a^2)$

b)ii)

Equation of normal:

$$m_{NORMAL} = -\frac{1}{4a}$$

$$y - 2a^2 = -\frac{1}{4a}(x - a)$$

When $x = 0$

$$y - 2a^2 = -\frac{1}{4a}(0 - a)$$

$$y = \frac{1}{4} + 2a^2$$

$$y = \frac{1 + 8a^2}{4}$$

Co-ordinates of N are $\left(0, \frac{1 + 8a^2}{4}\right)$

b)iii)

Area of $\triangle NTA$

If the base is NT

then the perpendicular height is a

$$Area = \frac{1}{2} \times NT \times a$$

$$Area = \frac{1}{2} \times \left(\frac{1+8a^2}{4} + 2a^2 \right) \times a$$

$$Area = \left(\frac{1+16a^2}{8} \right) \times a$$

$$Area = \frac{a+16a^3}{8} = \frac{a}{8} + 2a^3$$

b)iii)

First find the value of a so that $\triangle NTA$ is isosceles

Method1

$\triangle NTA$ is isosceles when

$$\angle NTA = \angle TNA = 45^\circ$$

If $\angle NTA = 45^\circ$ then $m_{AT} = \tan 45^\circ = 1$

$$m_{AT} = 4a = 1$$

$$a = \frac{1}{4}$$

Method2

$\triangle NTA$ is isosceles when

$$AT = AN$$

$$a^2 + \left(\frac{1+2a^2}{4} - 2a \right)^2 = a^2 + (-2a - 2a)^2$$

$$a^2 + \left(\frac{1}{4} \right)^2 = a^2 + (-4a)^2$$

$$a^2 + \frac{1}{16} = a^2 + 16a^4$$

$$\frac{1}{16} = 16a^4$$

$$\frac{1}{256} = a^4$$

$$a = \pm \frac{1}{4}$$

$$a = \frac{1}{4}$$

Method3

$\triangle NTA$ is isosceles when

y -value of $A(a, 2a^2)$ is the midpoint of NT

$$2a^2 = \frac{1}{2} \left(\frac{1+8a^2}{4} - 2a^2 \right)$$

$$4a^2 = \frac{1+8a^2}{4} - 2a^2$$

$$16a^2 = 1+8a^2 - 8a^2$$

$$a^2 = \frac{1}{16}$$

$$a = \pm \frac{1}{4}$$

$$a = \frac{1}{4}$$

$$\begin{aligned} Area of \triangle NTA &= \frac{a}{8} + 2a^3 \\ &= \frac{1}{4} \times \frac{1}{8} + 2 \left(\frac{1}{4} \right)^3 \\ &= \frac{1}{32} + \frac{1}{32} = \frac{1}{16} \end{aligned}$$